Customer Shopping Behavior Analysis

Uncovering insights from transactional data to guide strategic business decisions.

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Project Overview

1

Analyze Data

3,900 purchases across categories.

2

Uncover Insights

Spending patterns, segments, preferences.

3

Guide Decisions

Optimize business strategy.

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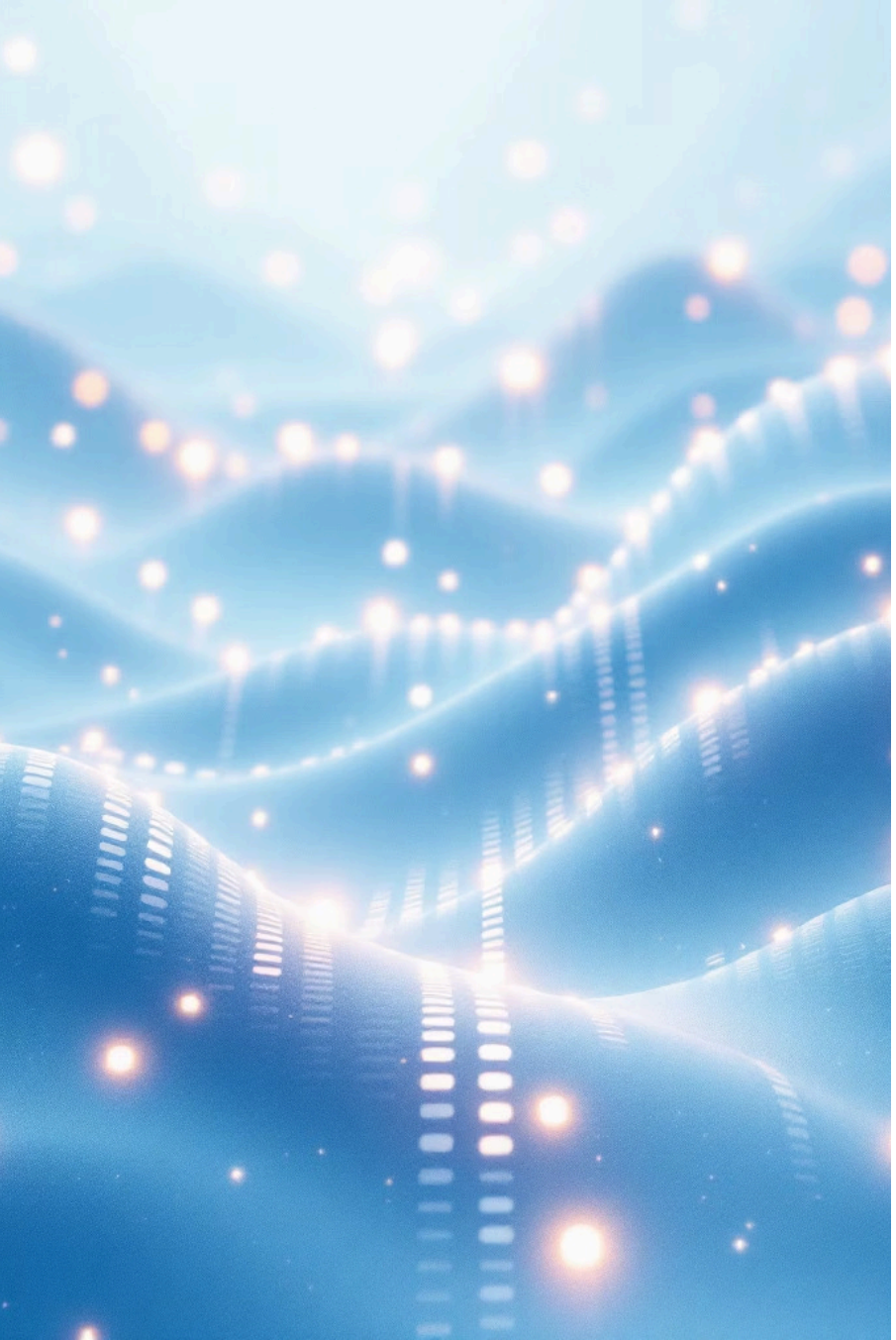

Dataset Summary

Key Features

- Customer demographics
- Purchase details
- Shopping behavior

Data Points

- 3,900 Rows
- 18 Columns
- 37 Missing values (Review Rating)

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Exploratory Data Analysis (Python)

01

Data Loading

Imported dataset using `pandas`.

02

Initial Exploration

Checked structure and summary statistics.

03

Missing Data Handling

Imputed `Review Rating` using median.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days`.

05

Database Integration

Loaded cleaned data to PostgreSQL.

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Data Analysis (SQL)

Structured analysis in PostgreSQL for key business questions.

1

Revenue by Gender

Male: $157,890, Female: $75,191

2

High-Spending Discount Users

839 customers identified.

3

Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, Skirt.

4

Shipping Type Comparison

Express: $60.48, Standard: $58.46

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SQL Analysis: Key Insights

1

Subscribers vs. Non-Subscribers

Non-subscribers: 2847 customers, avg spend $59.87, total revenue $170,436.

2

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%).

3

Customer Segmentation

Loyal (3116), Returning (701), New (83).

4

Repeat Buyers & Subscriptions

958 repeat buyers are subscribers.

5

Revenue by Age Group

Young Adult: $62,143, Middle-aged: $59,197.

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Top Products by Category

Rank	Category	Item Purchased
1	Accessories	Jewelry
1	Clothing	Blouse
1	Footwear	Sandals
1	Outerwear	Jacket

Most purchased items within each category.

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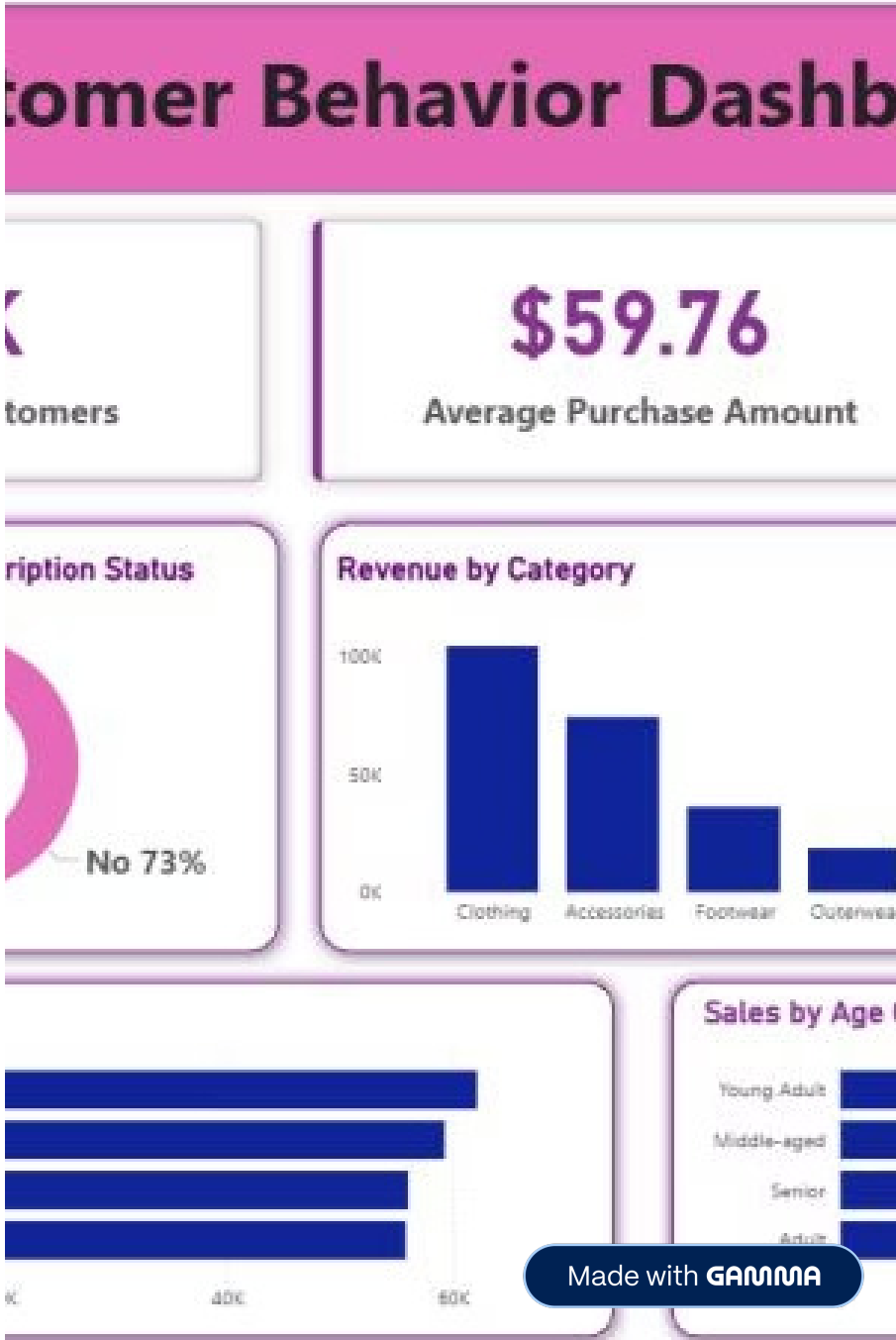
Power BI Dashboard

Interactive dashboard for visual insights.



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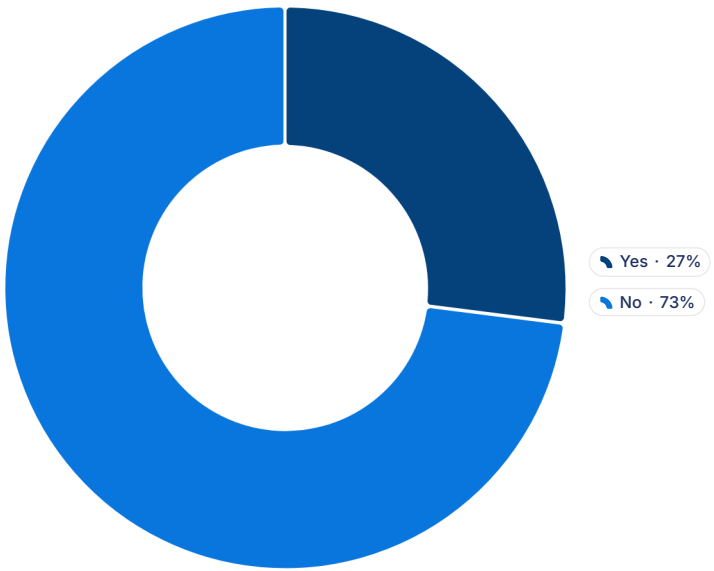
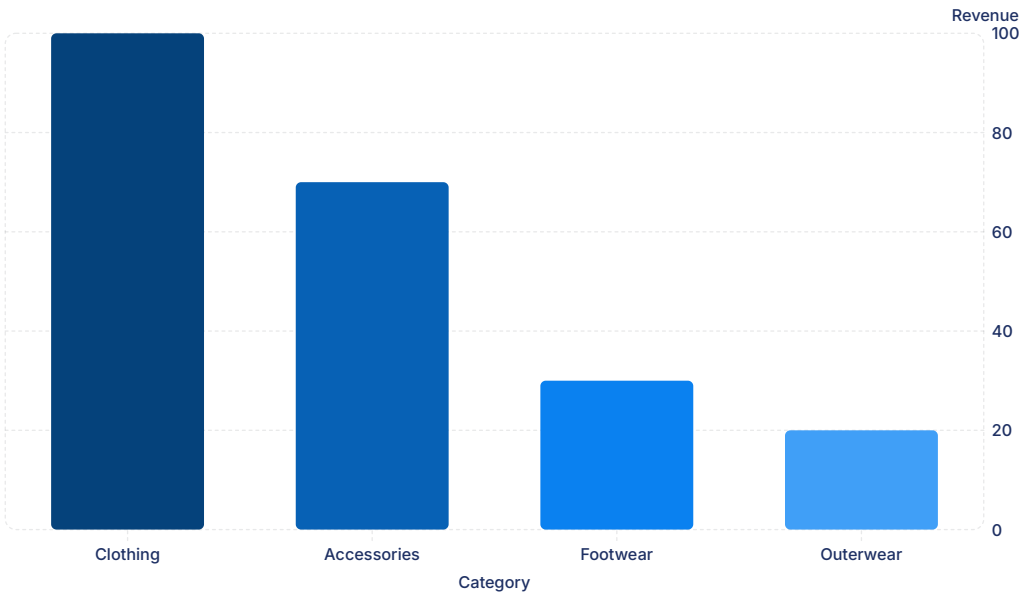
Dashboard Highlights



27% of customers are subscribers.



Clothing leads in revenue.

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits.

Customer Loyalty

Reward repeat buyers.

Review Discount Policy

Balance sales with margin.

Product Positioning

Highlight top-rated items.

Targeted Marketing

Focus on high-revenue groups.

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